Degrees of Freedom

Probability & Statistics – notes for regression and t-tests

Definition

Degrees of freedom (df) is the number of independent pieces of information that are still free to vary after some of the data has been used to estimate parameters.

1. Simple example: the mean

Take three numbers x_1, x_2, x_3 whose mean is fixed at 10, so

$$x_1 + x_2 + x_3 = 30$$

If $x_1 = 8$ and $x_2 = 12$ are chosen freely, x_3 is forced to be 10. Only 2 of the 3 values were free. This is why the sample variance uses

$$df = n - 1$$

One degree of freedom is used up by estimating the mean.

2. Regression example

Data set with n = 8 rows and the model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + u$$

Three parameters are estimated: β_0 (intercept), β_1 and β_2 (slopes), so k = 3.

$$df = n - k = 8 - 3 = 5$$

The model spends 3 pieces of information on the coefficients, leaving 5 independent pieces to estimate the error variance.

3. Why df matters for the t-test

Residuals are $e_i = Y_i - \hat{Y}_i$ (actual minus predicted value). Because the same data was used to estimate the betas, the residuals are not all free, so the error variance is found by dividing by n − k instead of n:

$$s^2 = \Sigma\, e_i^2 \,/\, (n - k)$$

The value $df = n - k$ is then used for:

- the t-test on each coefficient
- the p-value
- confidence intervals for the coefficients

Summary

Case	Degrees of freedom
Sample variance	$n - 1$
Regression with k estimated betas (including β_0)	$n - k$
Example above ($n = 8$, $k = 3$)	5

Memory aid: start with n pieces of information and subtract one for every parameter estimated from the data.